Pre-Class Assignment: Chapter 6

Automation

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Gans, Chapter 6 (Automation), Sections 6.1–6.3 and Section 6.4.3 (the AI taxonomy table). Sections 6.4.1–6.4.2 and 6.5 are optional.

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

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1. Gans gives a precise definition of **automation** in Section 6.2. In your own words, what does it mean for a decision to be automated?
2. Why does Gans insist that **AI adoption is not the same as automation**? Give a brief example of AI adoption *without* automation.
3. In Section 6.1, Gans argues that the standard task-based approach (Acemoglu and Autor) is not the right unit of analysis for studying AI. What unit does he propose instead, and why?
4. The human's expected payoff in the same model is $\delta(e + (1 - e)\alpha)R$. Briefly explain what each of the three parameters δ, e, and α represents.
5. Gans says humans have one **disadvantage** and one **advantage** relative to smart machines. State each in one sentence.
6. Figure 6.1 shows three regions in $(\alpha, C/R)$ space: "Rules or dumb machine," "Smart machines," and "Humans." Briefly describe what kind of task lands in each region.
7. Gans's main comparative-static result is: as AI prediction e improves, **smart machines substitute at the margin for human decision performance**. What is the intuition for why human advantage shrinks as e rises?
8. Table 6.1 in Section 6.4.3 distinguishes four types of AI: **Replacement**, **Augmentation**, **Unreliable**, and **Antagonistic**. Pick *one* of the two right-column types (Unreliable or Antagonistic) and explain in one or two sentences why a principal might deliberately prefer a *worse* AI in that case.
9. In one sentence, state the main takeaway of Chapter 6.